LANCERSNAP

Software Design Document (SDD)

CSCI 5520 - AI-Driven Software Development

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1 Problem & Users

1.1 Problem Description

Students at California Baptist University encounter physical and operational problems across campus, including broken equipment, facility issues, damaged furniture, maintenance problems, housing issues, dining-area problems, and other campus conditions that require staff attention.

Although CBU already has staff and departments responsible for these issues, students may not always know **who is responsible for a particular problem or how it should be reported**.

Key problems include:

- **Unclear reporting responsibility**

  Students may not know whether an issue should be reported to Housing, Facilities, Dining, building administration, IT, or another campus team. This can result in reports being sent to the wrong person or not being reported at all.

- **Incomplete or inconsistent information**

  Students may describe an issue differently or omit important information such as the location, type of problem, or severity. Staff may therefore spend additional time determining what the issue is and where it needs to be handled.

- **Fragmented reporting channels**

  Reports may currently originate through different people and processes, such as RAs, building reception staff, dining employees, or other campus personnel. This makes it difficult to maintain a centralized view of recurring campus issues.

- **Duplicate reports**

  Multiple students may report the same problem independently. Without duplicate detection, staff may receive several reports for one physical issue.

- **Limited visibility after reporting**

  Students may not know whether their report was received, assigned, or resolved. Staff may also lack a centralized view of issue volume, recurring problems, and resolution status.

- **Operational prioritization**

  Routine maintenance issues and potentially urgent hazards should not necessarily be treated with the same priority. Staff need structured information to help identify issues requiring faster attention.

Core problem

CBU has existing personnel and departments capable of handling campus issues, but students lack a simple, centralized reporting layer that captures useful information and routes the issue to the appropriate CBU team.

LancerSnap addresses this problem by allowing a student to **take a picture, provide a location, and submit a report without needing to determine which department is responsible.**

1.2 Target Users & Stakeholders

2 Proposed Solution & Scope

User / Stakeholder	Relationship to the Problem
CBU Student	

	Primary user who needs a fast and simple way to report a campus issue without determining the responsible department.
Campus Staff	Reviews, assigns, updates, and resolves reports routed to their team.
Housing / RAs	Receives housing-related reports and handles issues within their assigned responsibilities.
Facilities	Handles physical campus maintenance, infrastructure, equipment, and facility-related issues.
Dining Staff	Handles issues occurring in dining facilities and food-service areas.
Building Administration	Handles reports associated with a specific building or building-level operations.
IT / Technology Support	Handles technology-related campus issues when included in the routing configuration.
LancerSnap Administrator	Manages users, categories, routing rules, priorities, and system configuration.

2.1 Proposed Solution

LancerSnap is an **AI-assisted campus reporting and issue-routing application for California Baptist University**.

The core experience is designed around:

See an issue → Take a picture → Add location → Submit → CBU handles the rest.

Students do not need to determine which department, staff member, or office is responsible for the problem.

A student can capture or upload a photograph, provide an optional description, and confirm the location. The system then uses AI to analyze the report and generate recommendations such as:

- issue category
- extracted details
- possible urgency
- potential duplicate reports
- suggested routing destination

The student reviews the AI-generated interpretation before submitting the report.

After submission, LancerSnap applies configurable routing rules based on information such as **issue category and location** and sends the report to the appropriate CBU team.

The responsible staff member can then:

1. Review the report.
2. Review AI-generated information.
3. Assign or accept the issue.
4. Update its status.
5. Resolve the issue.
6. Notify the student.

Students can track their reports from submission through resolution.

The system is therefore not intended to replace CBU's existing RAs, Facilities staff, Dining staff, reception personnel, or other campus employees.

Instead:

LancerSnap is the centralized intake and tracking layer connecting students to existing CBU operational teams.

2.2 Minimum Viable Product (MVP) Scope

Student capabilities

Students can:

- Sign in using an authorized CBU account.
- Take or upload a photograph.
- Add an optional description.
- Select or confirm a CBU building or location.
- Submit a report without selecting the responsible campus office.
- Review AI-generated classification before submission.
- Track submitted reports.
- Receive notifications when report status changes.
- Reopen a report when the reported issue has not actually been resolved.

Supported report categories

The MVP will focus primarily on operational and physical campus issues such as:

- Facilities issue
- Housing issue
- Dining issue
- Building / common-area issue
- IT / technology issue
- Lost item
- Found item
- Safety-related campus issue
- Other

Lost-and-found is retained as a supported report category, but it is **not the primary product workflow**.

The primary purpose of LancerSnap is campus issue reporting and routing.

AI Capabilities

The AI processing layer may:

- Analyze uploaded images.
- Analyze optional student descriptions.
- Identify the likely issue category.
- Extract useful visual details.
- Suggest an urgency level.
- Suggest a routing destination.
- Identify potentially duplicate reports.
- Generate a concise report summary.

AI-generated information must be presented as a **recommendation**, not as an unquestionable fact.

The student and authorized CBU staff remain responsible for confirming the report information.

Staff Capabilities

Authorized CBU staff can:

- View reports assigned to their team.

- Review submitted photographs.
- Review student-provided descriptions.
- Review AI-generated classifications.
- Review suggested priority.
- Review duplicate-report information.
- Assign or accept reports.
- Update report status.
- Mark reports as resolved.
- Reopen reports when additional action is required.

Administrative Capabilities

Authorized administrators can:

- Manage report categories.
- Configure category-to-team routing rules.
- Configure location-based routing.
- Configure report priorities.
- Manage authorized staff access.
- Review aggregated report trends.
- View recurring issues by category or location.

Routing should be based on **teams or organizational roles rather than individual employees** wherever possible.

This prevents the system from becoming dependent on a specific RA or staff member when personnel assignments change.

2.3 Reporting Workflow

The primary LancerSnap workflow is:

Student → Photo → Location → AI Analysis → Duplicate Check → Routing → Staff Action → Resolution → Student Confirmation

Step 1 — Student reports an issue

The student opens LancerSnap and takes or uploads a photograph.

Step 2 — Location is provided

The student uses the current location or selects the appropriate CBU building/area.

Step 3 — AI analyzes the report

The system identifies the likely category, extracts useful details, and suggests a priority.

Step 4 — Student confirms

The student reviews the AI-generated information and can correct it before submission.

Step 5 — Duplicate detection

The system checks whether a similar unresolved issue already exists in the same area.

If a duplicate exists, the new report can be associated with the existing issue rather than creating unnecessary duplicate work.

Step 6 — Smart routing

The backend applies configurable routing rules using information such as:

Location + Category + Priority

The report is sent to the appropriate CBU team.

Examples include:

- Housing → Housing / RA
- Dining location → Dining team
- Building maintenance → Facilities
- Technology issue → IT
- Building-specific issue → Building administration

Step 7 — Staff action

The responsible team reviews the report, investigates the issue, and performs the necessary action.

Step 8 — Resolution

Staff updates the report to indicate that the issue has been resolved.

Step 9 — Student confirmation

The student receives a notification and can confirm that the issue has been resolved.

If the issue remains unresolved, the student can reopen the report.

2.4 Lost-and-Found Boundary

Lost-and-found will not be treated as a complete property-management system.

LancerSnap may allow students to report:

- Lost items
- Found items

However, LancerSnap will **not automatically determine ownership, physically transfer property, or require a finder to personally maintain possession of valuable property.**

For valuable or disputed items, the report should be routed to the appropriate CBU personnel or existing campus process.

The system may provide AI-assisted matching, but:

A similarity score does not establish ownership.

Any ownership decision remains subject to authorized human review.

2.5 Safety Boundary

LancerSnap may identify a report as a potential safety-related issue and assign a higher priority.

However:

LancerSnap is not an emergency response system.

The application must clearly communicate that immediate emergencies should be handled through CBU's established emergency procedures and appropriate emergency services.

LancerSnap does not directly dispatch police, fire, EMS, or other emergency services.

2.6 Out of Scope / Non-Goals

The following are intentionally outside the MVP:

- Harassment reports
- Disciplinary complaints
- Sensitive personal complaints
- Anonymous misconduct reporting
- Automatic disciplinary action
- Direct integration with 911, police dispatch, fire/EMS, or public emergency services
- Replacing CBU's existing work-order or case-management systems
- Replacing RAs, Facilities staff, Dining staff, or building administrators
- AI making final decisions about safety or administrative actions
- Facial recognition
- Identifying individual people from photographs
- Continuous GPS tracking
- Automatically transferring physical lost-and-found property
- Automatic ownership decisions for disputed items
- Public posting of submitted photographs
- Native desktop applications
- Fully automated resolution of ambiguous reports

Core non-goal

LancerSnap does not replace CBU personnel.

It provides:

Capture → Understand → Route → Track → Analyze

CBU personnel remain responsible for:

Review → Decide → Act → Resolve

3 User Stories & Acceptance Criteria

3.1 Student
US-01 — Submit Report

User Story:
As a student, I can take a picture, add a location and optional description, and submit a campus issue without knowing which office is responsible.
Pass/Fail Check:
Pass if the report is submitted and a confirmation is displayed.
US-02 — AI Classify Report
User Story:
As a student, I can review the category suggested for my report before submission.
Pass/Fail Check:
Pass if the system generates a supported category and allows the student to confirm or modify it.
US-03 — Track Report
User Story:
As a student, I can view the current status of my report.
Pass/Fail Check:
Pass if the current status is displayed and updated after an authorized change.
US-04 — Receive Resolution Update
User Story:
As a student, I am notified when staff update or resolve my report.
Pass/Fail Check:
Pass if the student receives the configured status notification.
US-05 — Reopen Report
User Story:
As a student, I can indicate that an issue marked resolved is still not fixed.
Pass/Fail Check:
Pass if the report can return to an active status.
US-06 — Report Location
User Story:
As a student, I can provide the building or campus location where the issue occurred.
Pass/Fail Check:
Pass if the submitted report contains a valid supported CBU location.

3.2 Campus Staff
US-07 — View Assigned Reports
User Story:
As campus staff, I can view reports assigned to my authorized team.
Pass/Fail Check:
Pass if staff can access only reports permitted by their role.
US-08 — Review AI Information
User Story:
As campus staff, I can review AI-generated classification, extracted details, and suggested priority.
Pass/Fail Check:
Pass if AI-generated information is visible within the report.
US-09 — Update Report Status
User Story:
As campus staff, I can update the status of a report.
Pass/Fail Check:
Pass if the updated status is saved and visible to the student.
US-10 — Resolve Report
User Story:
As campus staff, I can mark a report as resolved after taking appropriate action.

Pass/Fail Check:
Pass if the report moves to the configured resolved state and the student is notified.

3.3 Administrator

US-11 — Manage Routing
User Story:
As an administrator, I can configure category and location routing rules so that reports reach the appropriate team.
Pass/Fail Check:
Pass if a saved routing rule correctly determines the destination team.
US-12 — View Report Trends
User Story:
As an administrator, I can view aggregated reports by category, location, and time period.
Pass/Fail Check:
Pass if reports can be grouped by supported dimensions.
US-13 — Manage Categories
User Story:
As an administrator, I can configure supported report categories.
Pass/Fail Check:
Pass if an authorized administrator can create, modify, or deactivate a supported category.

4 Requirements, Constraints & Design Rules

RC-01 — Access Control

Students may access their own reports. Staff may access reports assigned to their authorized team. Administrative functions require administrator privileges.

RC-02 — AI Uncertainty

AI-generated categories, extracted information, urgency levels, and duplicate suggestions must be treated as recommendations when uncertainty exists.

RC-03 — Human Oversight

Authorized CBU staff remain responsible for final operational decisions and issue resolution.

RC-04 — Configurable Routing

Routing rules must be configurable by an authorized administrator rather than permanently embedded in the student interface.

RC-05 — Team-Based Routing

Reports should be routed to a responsible CBU team or organizational role rather than depending on a specific individual wherever possible.

RC-06 — Duplicate Handling

The system should identify potentially duplicate reports based on information such as location, category, time, and report similarity.

Duplicate detection must be treated as a recommendation rather than a guaranteed determination.

RC-07 — Safety Priority

Potential safety-related reports should receive a clearly identifiable higher priority.

RC-08 — Emergency Boundary

LancerSnap must clearly communicate that it does not replace emergency services or CBU emergency procedures.

RC-09 — Secure Media

Uploaded images must be stored using access controls and must not be publicly accessible by default.

RC-10 — Error Handling

If AI processing, image analysis, routing, or notifications fail, the system must clearly indicate the failure and must not report a false successful result.

RC-11 — Accessibility

Core reporting and tracking functions must use readable text, clear labels, sufficient contrast, and must not rely on color alone to communicate important information.

RC-12 — Status Transparency

Students must be able to understand the current state of their submitted report.

5 Platform & Architecture

5.1 System Components

LancerSnap will use a mobile-first architecture consisting of:

- Mobile application
- Backend API
- AI processing layer
- Routing engine
- Database
- Object/image storage
- Notification service
- Location services
- CBU staff dashboard

Recommended technology stack

Technology	Usage
React Native + Expo	Student and staff mobile interface
FastAPI / Python	Backend API and business logic
PostgreSQL	Structured application data
Supabase	Authentication, database access, and storage
Multimodal AI Model	Image analysis, classification, detail extraction, and priority suggestion
Vector / similarity search	Duplicate and similarity detection

Object Storage	Uploaded report photographs
Push Notifications	Status updates
Device / Campus Location	Building and location identification

5.2 Major System Components

Mobile Application

Provides:

- Photo capture
- Report creation
- Location selection
- AI result review
- Report tracking
- Notifications

Backend API

Handles:

- Authentication context
- Report creation
- Business rules
- Routing
- Permissions
- Status updates
- Communication with AI services

AI Processing Layer

Analyzes:

- Images
- Student descriptions
- Issue characteristics

Produces:

- Category suggestion
- Extracted details
- Priority recommendation
- Duplicate similarity information

Routing Engine

Determines the appropriate CBU team using configurable rules such as:

Category + Location + Priority

Database

Stores:

- Users
- Reports
- Categories
- Locations
- Routing rules
- Statuses
- Assignments
- Report history
- Duplicate relationships

Object Storage

Stores uploaded photographs separately from structured application records and applies access controls.

Notification Service

Provides:

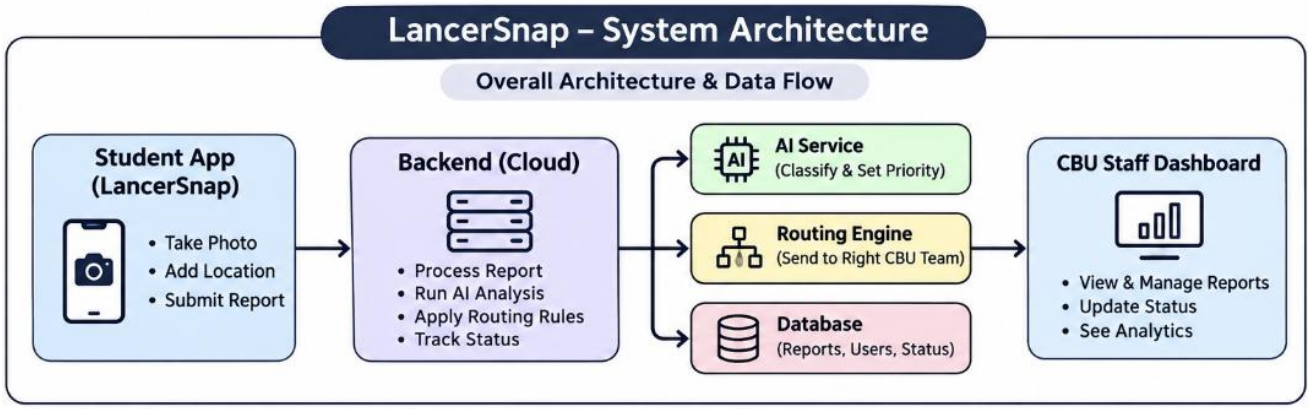
- Report submission confirmation
- Status updates
- Resolution notifications
- Important report notifications

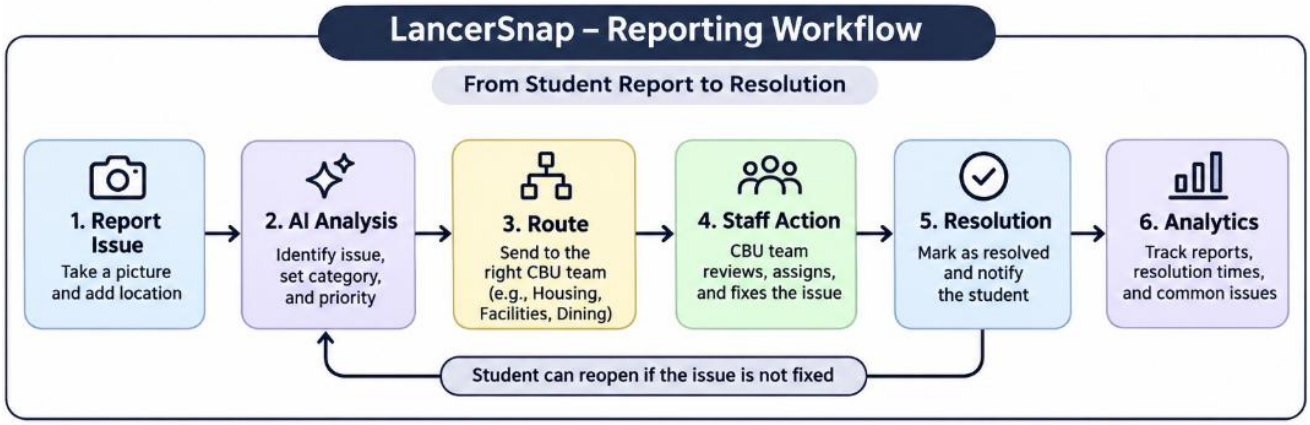
CBU Staff Dashboard

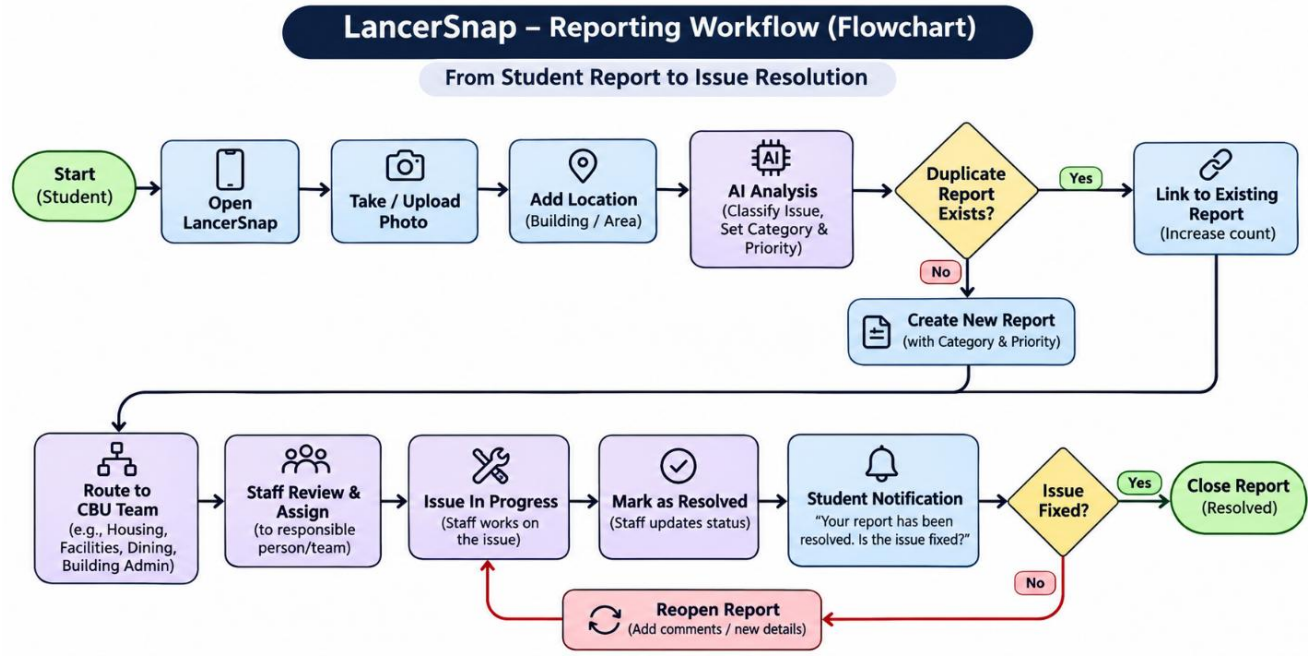
Allows authorized personnel to:

- View incoming reports
- Filter by priority/category/location
- Review AI information
- Assign reports
- Update status
- Resolve issues
- View operational trends

5.3 Architecture & Data Flow







6 UI/UX Design

6.1 Primary Application Flow

The primary student experience is:

Login

↓

Student Home

↓

Snap & Report

↓

Take / Upload Photo

↓

Add / Confirm Location

↓

AI Analysis

↓

Review & Confirm

↓

Submit

↓

Report Confirmation

↓

My Reports

↓

Status Timeline

↓

Resolved / Reopen

6.2 Staff Flow

Staff Login

↓

Staff Dashboard

↓

Report Queue

↓

Filter by Team / Category / Location / Priority

↓

Report Details

↓

Review Photo + AI Information

↓

Assign / Accept

↓

In Progress

↓

Resolve

↓

Student Notification

6.3 Duplicate Reporting Flow

When a student submits a report, the system checks for similar existing reports.

If no similar report exists:

Create New Issue

↓

Route to CBU Team

If a similar report exists:

Show Existing Issue

↓

Student Confirms Same Problem

↓

Link Report

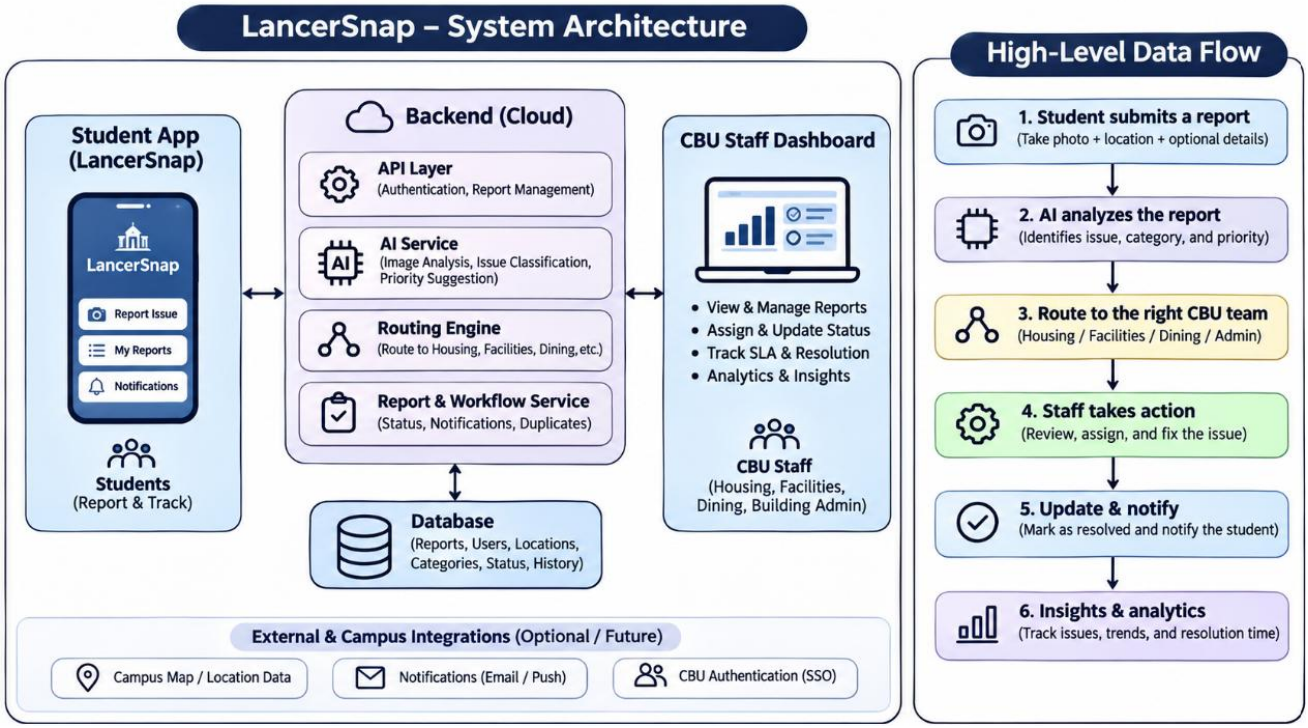
↓

Increase Affected-Student Count

↓

Existing Issue Continues Through Resolution

This reduces unnecessary duplicate work while preserving evidence that multiple students experienced the same problem.



Prototype / Figma Make Link:
https://www.figma.com/proto/makzxHRmehtu2jH9pJ4GZ6/Untitled?node-id=3-18121&p=f&t=6fP2TtLHqULakbnC-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1

7 Key Design Decisions & Open Questions

7.1 Key Design Decisions

DD-01 — Photo-First Reporting

LancerSnap uses a photo-first reporting process to minimize the effort required from students.

DD-02 — Centralized Intake

Students should not have to determine which CBU department is responsible for an issue.

DD-03 — AI-Assisted Classification

AI assists with classification, information extraction, duplicate detection, and priority suggestions.

AI does not make final operational decisions.

DD-04 — Smart Routing

Reports are routed using configurable category- and location-based rules.

DD-05 — Team-Based Routing

Routing targets organizational teams or roles rather than individual employees whenever possible.

DD-06 — Duplicate-Aware Reporting

Multiple students reporting the same physical issue should be able to contribute to a single operational issue rather than creating unnecessary duplicate work.

DD-07 — Transparent Status

Students should be able to see whether their report has been received, assigned, is in progress, or has been resolved.

DD-08 — Human Resolution

CBU staff remain responsible for investigating, fixing, and resolving reported issues.

DD-09 — Emergency Boundary

LancerSnap is a campus reporting tool, not an emergency response platform.

DD-10 — Lost-and-Found as a Secondary Category

Lost-and-found remains supported as a report category but is not the primary product workflow. Physical property handling and ownership decisions remain under appropriate CBU processes and authorized personnel.

7.2 Open Questions, Assumptions & Risks

OQ-01 — CBU Routing Rules

The exact mapping between categories/locations and responsible CBU teams needs to be confirmed.

OQ-02 — CBU Staff Integration

The final method for delivering reports to existing CBU teams needs to be determined.

OQ-03 — Notification Method

The final notification mechanism for student status updates needs to be selected.

OQ-04 — Location Accuracy

The project needs to determine whether campus building selection, GPS, or a combination provides the most reliable location information.

OQ-05 — AI Provider

The final AI model/provider should be evaluated based on accuracy, privacy, cost, and response time.

OQ-06 — Duplicate Detection

The final similarity method needs to be evaluated using representative campus issue reports.

Risks

RISK-01 — Incorrect AI Classification

AI may incorrectly identify the type of issue.

Mitigation: Allow student confirmation and staff review.

RISK-02 — Incorrect Routing

A report may be sent to the wrong team.

Mitigation: Use configurable routing rules and allow staff reassignment.

RISK-03 — Duplicate Detection Failure

Similar reports may not be recognized as duplicates.

Mitigation: Treat duplicate detection as a recommendation and allow staff to merge or associate reports.

RISK-04 — False Priority

AI may incorrectly classify the urgency of a report.

Mitigation: Priority remains advisory and staff retain final responsibility.

RISK-05 — AI Service Failure

External AI services may become unavailable or rate-limited.

Mitigation: Preserve the ability to submit a report even when AI processing fails, with the report entering a manual-review state where appropriate.

RISK-06 — Notification Failure

A student may not receive a status notification.

Mitigation: Maintain report status inside the application so notifications are not the only source of truth.

RISK-07 — Staff Adoption

Staff may not consistently use the system if it creates additional work.

Mitigation: Design the system around existing CBU teams and minimize duplicate data entry.

RISK-08 — Scope Expansion

Additional report categories could expand the system into sensitive, disciplinary, or emergency workflows.

Mitigation: Maintain a clear MVP boundary and require explicit design decisions before adding new report types.

8. Summary

LancerSnap is designed as an **AI-assisted campus reporting and routing platform for California Baptist University**.

Its central interaction is intentionally simple:

Take a picture → Add location → Submit.

The student does not need to determine who is responsible.

LancerSnap uses AI to structure the report, identify the likely issue category, suggest priority, detect potential duplicates, and assist with routing.

Existing CBU teams remain responsible for investigating and resolving the issue.

The system therefore creates a clear separation:

LancerSnap

Capture → Understand → Route → Track → Analyze

CBU

Review → Assign → Act → Resolve

The long-term value of the platform is not only faster reporting. Aggregated reports can help CBU identify recurring problems, locations with repeated issues, average resolution times, and areas where operational improvements may be needed.

Lost-and-found remains a secondary report category rather than the central identity of the product. This keeps the MVP focused on the problem LancerSnap is best positioned to solve: **making campus issue reporting simple for students and actionable for CBU staff.**

9 References & Supporting Links (Optional)

- California Baptist University - Official university website and campus information.
  https://calbaptist.edu/

- Figma - Figma Make documentation.
  https://help.figma.com/

- React Native - React Native documentation.
  https://reactnative.dev/

- FastAPI - FastAPI documentation.
  https://fastapi.tiangolo.com/

- Supabase - Supabase documentation.
  https://supabase.com/docs